



Vehicle Walay

AI-Powered Vehicle Discovery &
Recommendation Platform

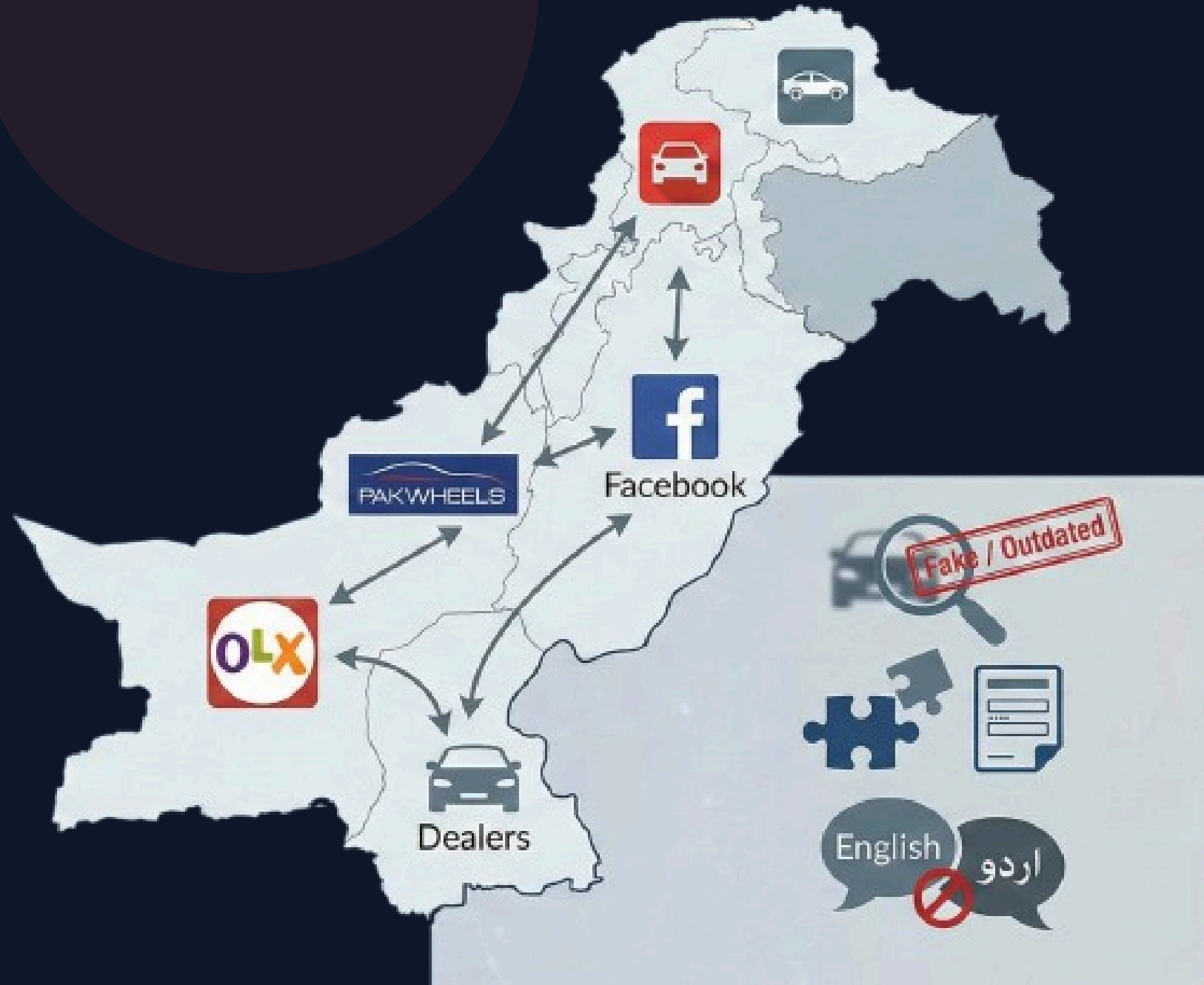
Group Members:

Bilal Aleem (221-1866)

Awais Khaleeq (221-2039)

Supervisor: Adil Majeed

FINAL YEAR PROJECT PROPOSAL DEFENSE



Problem Statement

Finding vehicles in Pakistan is frustrating and inefficient. Buyers have to browse PakWheels, OLX, Facebook groups, dealership sites, and WhatsApp forwards. Many listings are outdated, fake, already sold, or missing key details.

Most platforms assume English literacy. A large population searches in Urdu or Roman Urdu, which current systems do not understand properly.

Solution/Goals

VehicleWalay solves this with an AI-powered RAG chatbot that delivers personalized vehicle recommendations, market insights, and automated interactions for buyers and dealers.

-  Understands user queries via Natural Language Understanding (NLU).
-  Supports multilingual text and voice queries (English & Roman Urdu).
-  Automates tasks such as test-drive scheduling with dealers, AI-generated vehicle walkthrough videos, and on-demand PowerPoint presentations.

**VehicleWalay Assistant**
AI-Powered · Always Available

I need a 3-seater car in Islamabad with a budget of 15 lakh

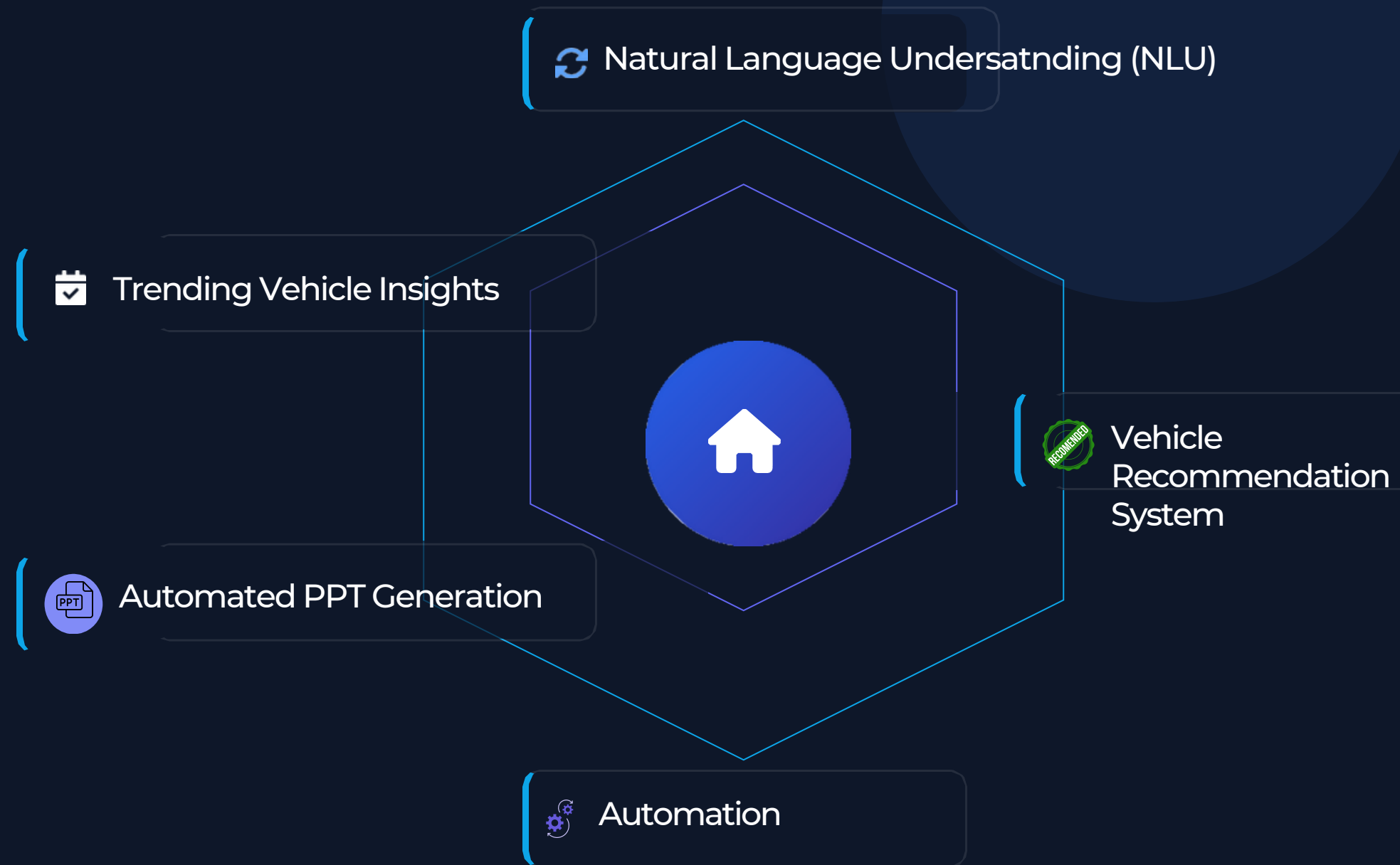
I found 5 vehicles matching your criteria in Islamabad. Here's the best match:

Would you like to view vehicle details or get more recommendations?

Type your message... 

Scope / Features

-  **Natural Language Understanding (NLU)**
Advanced AI understanding to interpret user vehicle queries and deliver accurate vehicle search results.
-  **AI-Powered vehicle Recommendation System**
Personalized vehicle suggestions based on user preferences such as budget, brand, fuel type, and usage.
-  **Automation (Alerts)**
Instant notifications when saved vehicles change price, get sold, or similar listings appear.
-  **Fake Vehicle Listing Detection**
Detects suspicious listings based on unusually low prices and fake or misleading descriptions..
-  **Automated Presentation (PPT) Generation**
One-click generation of vehicle presentations and spec sheets tailored for buyer or dealer meetings.



EXPANDING THE
POSSIBILITIES

Scope / Features

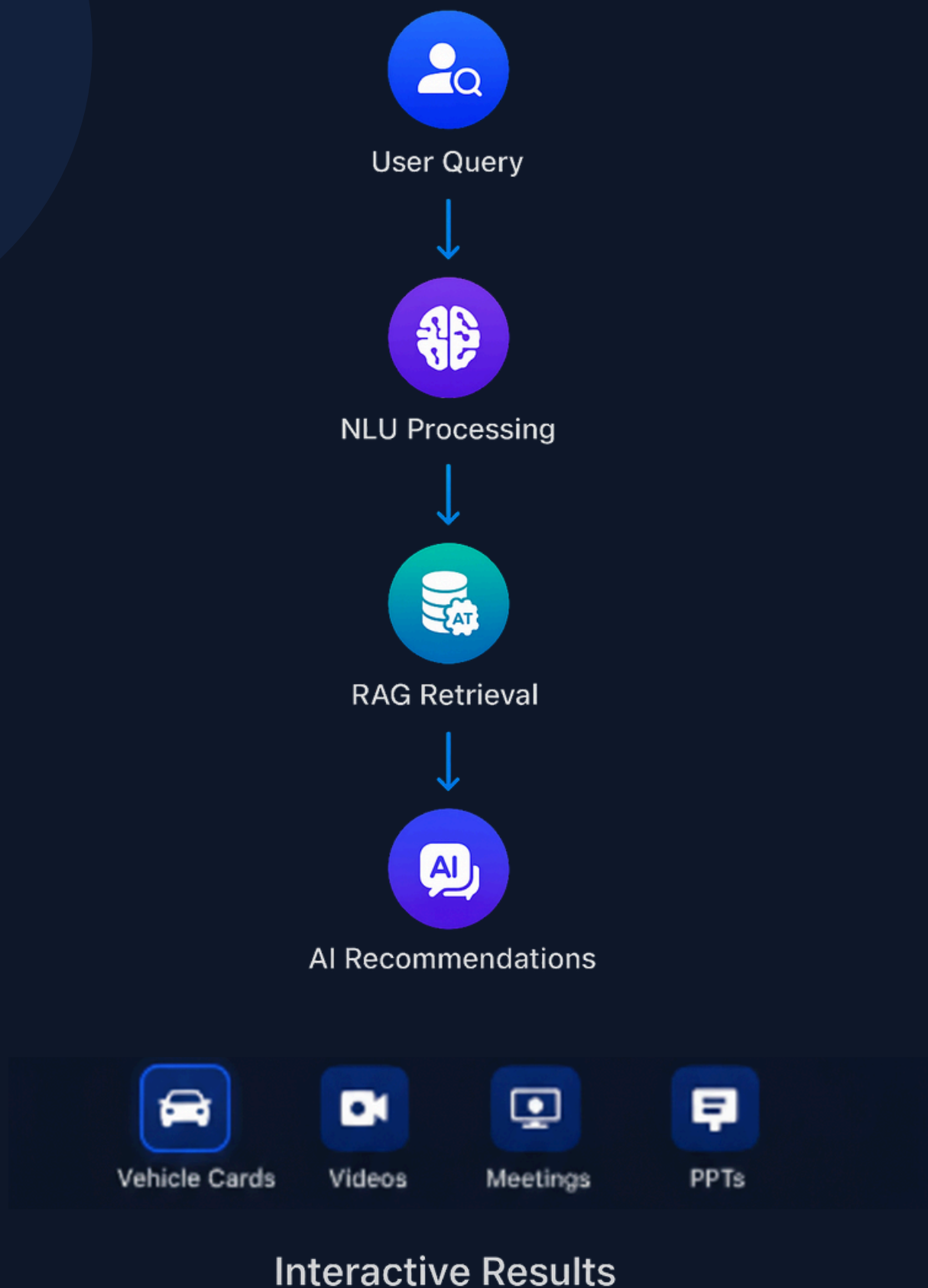
-  **Real-time Listings Integration**
Seamless integration with vehicle databases for up-to-date listings
-  **Multilingual Chat**
Support for English and Roman Urdu to serve local market needs
-  **Property Content Generation**
Automated creation of video videos
-  **buyer/seller Meeting Automation**
Streamlined scheduling and coordination between seller and buyers



EXPANDING THE
POSSIBILITIES

Existing Systems

Features	 PakWheels.com	 OLX.com.pk	 Facebook Groups	 Dealer Websites	 VehicleWalay
 Natural Language Understanding (NLU) AI-Powered Vehicle Recommendation					
 Real-Time Vehicle Listings Integration					
 Automation (Price Change/Sold Notifications)					
 Fake Vehicle Listing Detection					
 Multilingual Support (English & Roman Urdu)					
 Vehicle Video Generation					
 buyer/seller Meeting Automation					
 AI-Powered vehicle Recommendation System					



Process Flow

VehicleWalay's intelligent workflow processes user requests through multiple stages to deliver personalized vehicle results.



User Query

Natural language input in English or Roman Urdu



NLU Processing

Understanding user intent and requirements



RAG Retrieval

Finding relevant listings from multiple sources



AI Recommendations

Personalized vehicle suggestions



Interactive Results

Vehicle card, videos, meetings, and PPTs

Tools & Technologies



Python



FastAPI



LangChain



Cloud Hosting



DB



Web Scraping



Docker



Backend Development

Python, FastAPI, LangChain



Data Management

DB, Web Scraping (Zameen, Graana, Booking)



Frontend & User Interface

Website + Customized Chatbot UI



Deployment & Infrastructure

Docker containerization, Cloud hosting

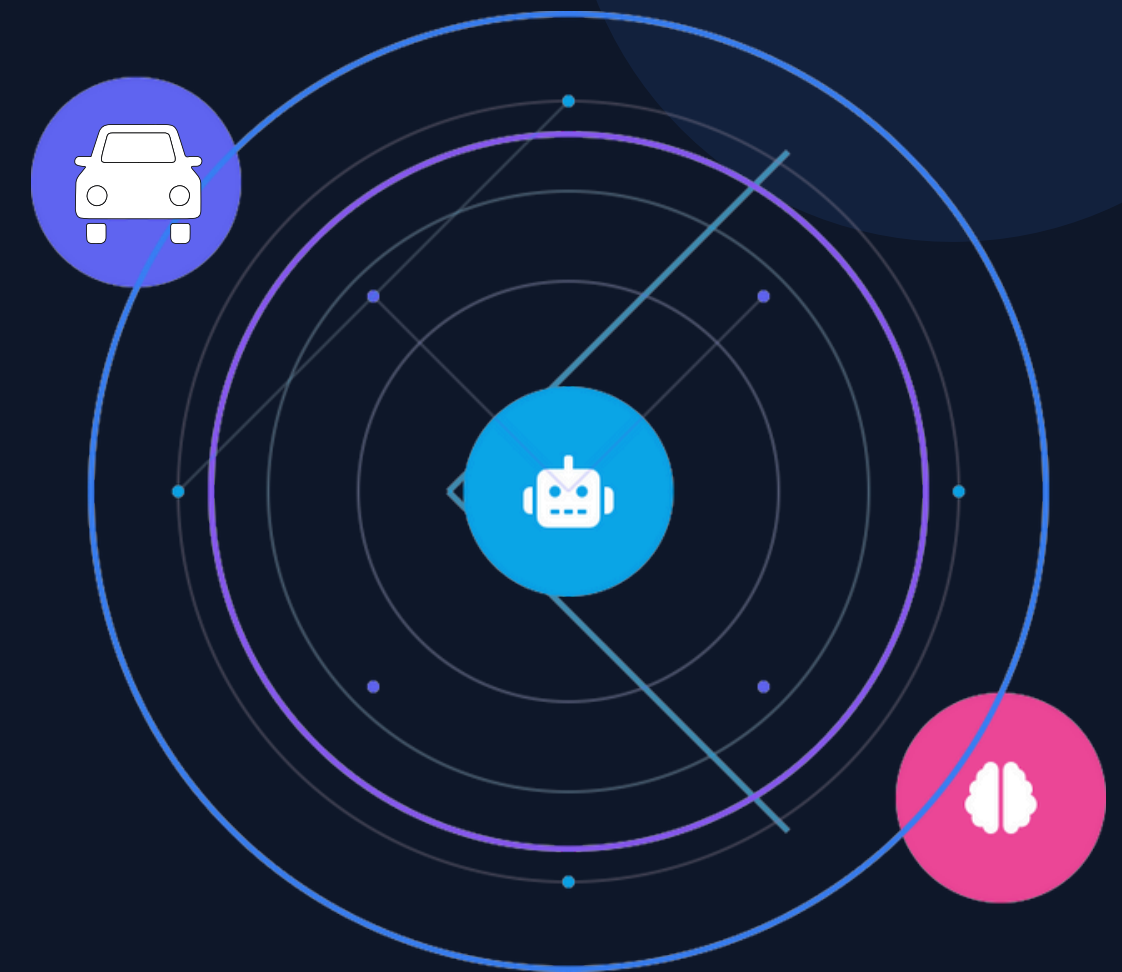
Work Division Timeline

Process	Feb-Mar	Apr-May	Sep-Oct	Nov-Dec
Real-Time Property Listings Integration	Bilal Aleem	Bilal Aleem		
Automation for Saved Properties	Awais Khaleeq			
Comprehensive Documentation	Awais Khaleeq	Bilal Aleem	Awais Khaleeq	Bilal Aleem
Trending Investment Areas Page		Awais Khaleeq		
AI-Powered Property Recommendation System		Bilal Aleem	Awais Khaleeq	
Automated PPT Generation		Bilal Aleem		
Multilingual + Voice Functionality			Bilal Aleem	
NLU Implementation			Awais Khaleeq	Bilal Aleem
Location-Based Video Generation				Awais Khaleeq
Meeting Automation with Agents				Awais Khaleeq
User History Management	Everyone	Everyone	Everyone	Everyone
Testing & Deployment	Everyone	Everyone	Everyone	Everyone

Awais Khaleeq
Bilal Aleem
Everyone

References

- ✓ <https://journalofbigdata.springeropen.com/articles/10.1186/s40537-022-00592-5>
- ✓ <https://www.olx.com.pk/motors/>
- ✓ <https://pypi.org/project/python-pptx/>
- ✓ <https://www.pakwheels.com/used-cars/>
- ✓ <https://www.pakwheels.com/>
- ✓ <https://en.wikipedia.org/wiki/PakWheels>
- ✓ <https://apps.apple.com/pk/app/pakwheels-buy-sell-cars/id739776365>



Thank You



Thank you for your attention

We're excited to revolutionize vehicle discovery in Pakistan with
VehicleWalay's AI-powered solutions



Find your dream Car with VehicleWalay!